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Report on the “Computer Vision in Archaeology” Training School in Brno, Czech Republic

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The training school was organized by the Archaeological Information System of the Czech Republic (AIS CR) research infrastructure at the Institute of Archaeology, Czech Academy of Sciences, Brno, together with the MAIA COST Action project.

During the 5-day workshop held in Brno, different datasets were used to showcase the possibilities that computer vision offers.

The first day was an introduction day, where we got an insight into the planned schedule for the week and an introduction to coding with agents and the use of Python for computer vision. Prior to the workshop, most of the data and the information related to the datasets and the necessary software were communicated to the participants, which made the preparation smoother.

On the second day, we focused on annotation tools. We tested CVAT, an open-source software. Using the image data provided by the school, we tested the strengths of such a computer vision application. Among the other tools that the organizers presented to us, and showed us where to find, a noteworthy one is OpenCV. The second half of the day was focused on the participants project presentations. Several interesting presentations were given, covering a variety of projects. The day finished with a nice tour of the city, which was a good addition to the classroom schedule and enabled us to learn more about the city of Brno.

Wednesday was dedicated to microscopic image datasets. During the morning, we learnt about some of the problems this workflow can have and how to deal with them. In the afternoon, we were able to test some examples of such images by applying different algorithms (CLAHE was used as an example). In the break between the theoretical and practical parts, we had an opportunity to visit the institute's laboratories and learn more about their work.

The following day was dynamic, with two different datasets explored. The morning session covered satellite imagery in archaeology and showed us how computer vision can help us work with this data. Going through the data and the exercise together, we learned how helpful computer vision can be for this kind of analysis.

In the afternoon, we switched to the rock art imagery workflow. We saw how machine learning together with computer vision can help with the detection and recognition of rock art.

The last day was an introduction to the coin dataset, the problems it involves, and the solutions we can apply. To close the week, we got a short introduction on where to find relevant information and some tips on how to use the Hugging Face site.

In conclusion, the week brought a lot of new information and taught me what computer vision can do. By using different datasets, we gained insight into different problems and research aspects. The organization of the workshop was well done and left us participants with a lot of data and material we can use in the future. The perfect balance of learning, training, and networking made the workshop a very rewarding experience. I went back home with good memories, enriched as a person and researcher.